GSD RESEARCH MISSION PACKAGE — FINAL

Drift in LLM Systems

Knowledge, Alignment, and Retrieval

Executed Mission — Four Modules,
Unified Taxonomy, CAPCOM-Gated Publication

Vision → Research → Mission → **Executed Artifact**

15 Primary arXiv Sources | 14 Supporting Sources | Four Modules

Unified Taxonomy | Test Plan (30+ IDs across SC/CF/IT/EC) | Verification Matrix

Milestone: v1.49.569 | Date: April 23, 2026 | Branch: dev

Phase 689 — W3 Publication | CAPCOM W3: Hard-Block Gate

Compile: `xelatex --shell-escape` (three passes + bibtex)

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STAGE 1 — VISION FRAMING (CONDENSED)

1 Drift as the Space Between

Date: April 23, 2026

Milestone: v1.49.569 — Drift in LLM Systems

Status: Executed artifact (Wave 3 Publication)

Depends on: 15 primary + 14 supporting arXiv sources; see References

1.1 One-Paragraph Summary

This mission studies *drift*: the space between the behaviour an LLM was instructed to exhibit and the behaviour it actually produces, measured over the trajectory of its work. The full executed package shipped here collects four modules — knowledge (A), alignment (B), retrieval (C), synthesis (D) — each grounded in published 2024–2026 arXiv literature and each audited for numeric-attribution integrity, citation uniqueness, and quote-length discipline via the hard-blocking CAPCOM gate defined in Phase 684 and invoked at the final page of this document.

1.2 The Three Drift Surfaces

```
USER INTENT (t=0) | v +-----------------+-----------------+ | | | v v v
+---------+ +---------+ +---------+ | WEIGHTS | | CONTEXT | | INDEX | |
(prior) | | (state) | | (SSoT) | +----+----+ +----+----+ +----+----+ | | |
knowledge alignment retrieval /factual /task /context drift drift drift |
| | v v v WHAT IS WHAT I WHAT THE TRUE? SHOULD SOURCE DO? SAYS?  | /  v /
+------> SYNTHESIS <---------+ (coupling effects)
```

1.3 What This Document Adds to the Mission Package

The earlier `drift-mission.tex` artifact delivered vision, research reference, and a wave plan. This final artifact is the *executed* mission package: four written modules, three supporting tables, a unified taxonomy folding cross-module double-counts into single rows, a test plan with test IDs spanning the four category shapes (safety-critical, core-functional, integration, edge), a verification matrix mapping every DRIFT-* success criterion onto at least one test ID, and a reference list rendered from `sources/index.bib` via BibTeX. The CAPCOM gate at the end of this document is a hard block — a non-zero exit aborts the build and prevents a flawed artifact from being published.

MODULE A — KNOWLEDGE & FACTUAL DRIFT (WAVE 1A)

2 Module A — Knowledge & Factual Drift

Module ID: A

Surface: Knowledge

Primary sources (from sources/index.bib): spataru2024sd, fastowski2024knowledge, wu2025natural, mir2025lsd, drift2026probe

Supporting sources: min2023factscore, manakul2023selfcheck, lindsey2025faithful

Author phase: 685 (Opus)

Editorial basis: Phase 684.1 Opus review — see sources/meta.json review_status fields

2.1 Scope of the Knowledge Surface

Knowledge drift names the class of failures in which a language model's factual output diverges from ground truth over the course of generation, prompt exposure, or representational depth, without any change to its parameters. Four signatures define the surface. The first is *intra-output drift*: a long-form completion begins with correct facts and degrades as generation continues. Spataru et al. formalise this as the *semantic drift (SD) score*, reporting an SD band of 0.77–0.79 across six frontier LLMs on a 500-biography FActScore corpus [Spataru et al., 2024]. The second is *belief-shift drift*: injected misinformation in the prompt context displaces the model's factual estimate, measurable through its uncertainty distribution [Fastowski and Kasneci, 2024]. The third is *temporal drift against pretraining*: reading passages evolve through natural human editing and model accuracy on those passages degrades even when the answer content remains verbatim present [Wu et al., 2025]. The fourth is *representational drift across depth*: hallucinated continuations travel through the transformer on a different trajectory from factual ones, visible to layer-wise probes before generation completes [Mir, 2025] [team, 2026].

These four signatures share a common diagnostic shape — each supplies a scalar or vector separating *correct-locus* outputs from *drift-locus* outputs with statistical significance — but they differ on the intervention surface they expose. Intra-output drift admits early-stopping and rerank-at-inference mitigations. Belief-shift drift admits prompt-hygiene and retrieval-provenance mitigations. Temporal drift admits recency-aware retrieval mitigations (handed off to Module C). And representational drift admits probe-based routing and late-layer-interpretability mitigations, with the strongest reported probe [team, 2026] adding less than 0.1% [team, 2026] overhead and gaining up to 13 AUROC points [team, 2026] over prior methods. The taxonomy rows and mitigation decision tree in §2.3 and §2.4 bind each signature to its metric and intervention branch.

One editorial note. A body of adjacent work on *intermediate-representation faithfulness* — Lindsey, Chen et al. 2025 — is motivational for the layer-wise probe family.

Phase 684.1 flagged this reference as `review_status: partial` (prose mention in `drift-mission.tex` without a canonical arXiv ID). This module cites the reference by [Lindsey and et al., 2025] only to credit the motivating intuition, and surfaces no numerical claim attributed to it. All quantitative headlines in the findings below attach to papers with `review_status: supported`.

2.2 Cited Findings

2.2.1 Semantic Drift in Long-Form Generation

Spataru, Hambro, Voita, and Cancedda introduce the *semantic drift score*: a scalar quantifying the separation between correct and incorrect atomic facts in a generated paragraph relative to a FActScore-scored ground truth [Spataru et al., 2024]. Evaluating six frontier LLMs — Falcon-7B (77% [Spataru et al., 2024]), Falcon-40B (77% [Spataru et al., 2024]), text-davinci-003 (77% [Spataru et al., 2024]), GPT-3.5 (79% [Spataru et al., 2024]), GPT-4 (78% [Spataru et al., 2024]), LLaMA-2-70B (78% [Spataru et al., 2024]) — they find an SD score that clusters tightly in the 0.77–0.79 [Spataru et al., 2024] band on a 500-biography FActScore corpus (random-permutation p under 0.02 [Spataru et al., 2024]). The drift point arrives early: in 37% [Spataru et al., 2024] of paragraphs the drift falls within the first 10% [Spataru et al., 2024] of facts, and only 7% [Spataru et al., 2024] of paragraphs contain at least ten correct facts before the first incorrect one.

The paper's mitigation tradeoff curve is the strongest single piece of knowledge-drift evidence in the corpus. An oracle early-stop at the drift point raises FActScore from 44.6% [Spataru et al., 2024] to 81.7% [Spataru et al., 2024], eliminating 92% [Spataru et al., 2024] of incorrect facts at the cost of 58% [Spataru et al., 2024] of the correct ones. Practical stopping heuristics recover a substantial fraction of the oracle gain: encouraging the EOS token when it falls in top-k yields 70% [Spataru et al., 2024] factuality; using SelfCheck-BERTScore [et al., 2023a] as an early-stop signal produces a tradeoff curve anchored at 70%+ [Spataru et al., 2024] accuracy; a resample-then-rerank pipeline with SC-BERTScore lifts factuality by 8.7 pp [Spataru et al., 2024] (to 53.3% [Spataru et al., 2024]) and composes additively with early stopping.

2.2.2 Knowledge Drift via Misinformation

Fastowski and Kasneci define *knowledge drift* as the displacement of an LLM's factual belief state under prompt-time exposure to false information [Fastowski and Kasneci, 2024]. Evaluating GPT-4o, GPT-3.5, LLaMA-2-13B, and Mistral-7B on QA tasks under three prompt conditions (clean, false-information injection, random-information injection) and tracking accuracy alongside three uncertainty metrics (entropy, perplexity, token probability), they isolate a two-phase dynamic. On *first* exposure to false information, uncertainty rises by up to 56.6% [Fastowski and Kasneci, 2024] as the model wavers between the injected claim and its parametric prior. On *repeated* exposure, uncertainty drops 52.8% [Fastowski and Kasneci, 2024] below the untainted baseline — the model has internalised the lie and become more confident in it than in truth. The signature is visible across all four evaluated models, with open-source code at `github.com/afastowski/knowledge_drift`.

The editorial review assigns this finding `rigor_score: 4` [Fastowski and Kasneci, 2024], noting that accuracy-versus-uncertainty decoupling is a single-paper result

that warrants replication; the four-model panel is adequate but not exhaustive.

2.2.3 Natural Context Drift Against Pretraining

Wu, Schlegel, and Batista-Navarro evaluate how LLM QA accuracy degrades as reading passages evolve via natural human editing relative to the versions those passages held at pretraining time [Wu et al., 2025]. Across six QA datasets (including BoolQ) and eight LLMs with public training data, they compute paragraph-level semantic similarity between the edited passage and the pretraining-era version. Average BoolQ accuracy drops by more than 30 pp [Wu et al., 2025] between the highest- and lowest-similarity bins; regression slopes exceed 70 [Wu et al., 2025] across several LLMs. Human annotators do *not* show this degradation: the answer content remains present in the edited passage, so the drop indexes reliance on pretraining surface form rather than semantic comprehension [Wu et al., 2025].

The result doubles as a retrieval-drift signal that Module C inherits: a recency-aware retrieval strategy that rewrites pretraining-form passages toward their current-form variants should recover a bounded fraction of the 30 pp [Wu et al., 2025] gap. The review flags a title variance between extraction and bibliography rendering of this paper; both forms refer to the same arXiv:2509.01093 entry and the numerical finding is unaffected.

2.2.4 Layer-wise Geometric Hallucination Detection (LSD)

Mir introduces *Layer-wise Semantic Dynamics (LSD)*, a framework that analyses the evolution of hidden-state semantics across transformer layers to detect hallucination intrinsically [Mir, 2025]. Using margin-based contrastive learning, LSD aligns hidden activations with factual encoder embeddings and distinguishes a *stable-alignment* trajectory (factual completions) from a *late-layer-divergence* trajectory (hallucinated completions). Reported detection performance on TruthfulQA and synthetic factual-hallucination corpora is F1 = 0.92 [Mir, 2025], AUROC = 0.96 [Mir, 2025], and clustering accuracy = 0.89 [Mir, 2025]. The mechanistic interpretation — early layers encode syntax, middle layers integrate semantics, late layers refine factual alignment — locates the failure at late-layer refinement and thus identifies a concrete layer-index intervention target.

[Mir, 2025] receives `rigor_score: 4` in Phase 684.1 review. The single-author status and custom synthetic-hallucination corpus alongside TruthfulQA justify a rigor discount, and module prose frames the F1 = 0.92 [Mir, 2025] and AUROC = 0.96 [Mir, 2025] numbers as *reported on the evaluated corpus* rather than as benchmark-standard performance.

2.2.5 Representational Inconsistency Probe

[team, 2026] is a lightweight probe over intermediate-layer representations that detects hallucination before generation completes. Adding less than 0.1% [team, 2026] computational overhead and running in parallel with autoregressive decoding, the probe reaches SOTA AUROC on 10 of 12 [team, 2026] evaluation settings across four QA benchmarks and three LLM families, with gains up to 13 AUROC [team, 2026] points over prior methods. A router built on the probe sends confident queries to a default backbone and delegates uncertain ones to a stronger model — the first practical probe-based routing story in the knowledge-drift corpus.

The cited mechanism is motivated by prior interpretability work on intermediate representations encoding reasoning signals not always faithfully expressed in the final-layer output [Lindsey and et al., 2025]. Because [Lindsey and et al., 2025] carries `review_status: partial` (no canonical arXiv identifier at the time of the Phase 684.1 review), the Lindsey reference is used for motivation only; every quantitative attribution in this subsection is to [team, 2026] directly.

2.3 Taxonomy Rows

Phase 685 appends six Module-A rows to `schema/drift_taxonomy.json` (one per signature, plus a derived FActScore-decay row drawn from the Spataru oracle-stop evidence). The rows are populated per the 7-field record schema `{phenomenon, surface="knowledge", metric, typical_magnitude, citation_id, mitigations, couplings}` declared in `schema/drift_taxonomy.json` and locked by `drift-mission.tex` §3.3. See that JSON file directly for the machine-readable form; the prose summary appears in Table 1.

Phenomenon	Signal	Typical Magnitude	Source
Semantic drift in long-form generation	SD score on atomic facts	0.77–0.79 across six LLMs	[Spataru et al., 2024]
Late-generation factuality decay	FActScore vs. stopping position	Baseline 44.6%; oracle-stop 81.7% on LLaMA-2-70B	[Spataru et al., 2024]
Misinformation-induced belief shift	Uncertainty deltas (entropy / perplexity / token-prob)	+56.6% first exposure; -52.8% repeated exposure	[Fastowski and Kasneci, 2024]
Natural context drift vs. pretraining	BoolQ accuracy vs. semantic-similarity bin	30+ pp accuracy drop across similarity extremes	[Wu et al., 2025]
Cross-depth semantic trajectory (LSD)	Hidden-state alignment vs. factual manifold	F1 = 0.92, AUROC = 0.96 on evaluated corpus	[Mir, 2025]
Intermediate-layer uncertainty probe	Layerwise-probe AUROC vs. final-layer detectors	+13 AUROC points over prior SOTA; <0.1% overhead	[team, 2026]

Table 1: Module A — Knowledge & Factual Drift taxonomy rows. Machine-readable form in `schema/drift_taxonomy.json`.

2.4 Mitigation Decision Tree

Given a detected knowledge-drift signal, Module A identifies four mitigation branches distinguished by detection locus, compute cost, and reported effect size. Table 2 renders the decision tree; the branch that fires is selected by the drift-signal shape (intra-output vs belief-shift vs representational) and by deployment constraints (single-pass generation vs resample-allowed vs router-available).

Branch	Trigger	Reported Effect	Compute Cost	Trade-off
Early-stop (oracle)	Intra-output drift; FActScore available	FActScore 44.6% → 81.7% on LLaMA-2-70B [Spataru et al., 2024]; removes 92% [Spataru et al., 2024] of incorrect facts	Negligible (truncate)	Removes 58% [Spataru et al., 2024] of correct facts — truncation is aggressive
Early-stop (heuristic)	Intra-output drift; no oracle available	70% [Spataru et al., 2024] factuality via top-k EOS; 70%+ [Spataru et al., 2024] via SC-BERTScore threshold [et al., 2023a]	Near-zero (decode-time)	Recovers a fraction of the oracle gain; noisy near drift point
Resample-and-rerank	Intra-output drift; latency budget available	+8.7 pp [Spataru et al., 2024] factuality (to 53.3% [Spataru et al., 2024]) via SC-BERTScore rerank; composes with early-stop	$N\times$ sampling cost	Additive with early-stop; N samples per query
Tool-calls / external QA	Intra-output drift; retrieval tool available	Near-zero improvement reported [Spataru et al., 2024] (Toolformer/Atlas baseline)	Retrieval round-trip cost	Model fails to gate retrieval calls against actual need — Spataru attribution
Probe-based router	Representational drift; late-layer probe available	+13 AUROC [team, 2026] over prior SOTA at <0.1% [team, 2026] overhead; SOTA on 10/12 [team, 2026] settings	<0.1% [team 2026]	Requires per-family probe training; delegation cost depends on traffic shape

Table 2: Module A — Knowledge-drift mitigation decision tree. Five branches spanning intra-output, belief-shift, and representational drift classes. Effect sizes are reported-on-benchmark; deployment gains will differ.

Two structural observations emerge. First, the *tool-call branch is the worst mitigator reported in the corpus* — Spataru et al. attribute the near-null improvement to the model's failure to gate retrieval against need [Spataru et al., 2024]. Module C (Retrieval & Grounding Drift) inherits this observation and addresses it with retrieval-discipline instruments. Second, the probe-based router branch is the only mitigator whose overhead is *sub-1%* of decode-time cost [team, 2026], making it the highest compute-efficiency intervention in the Module-A surface. These two observations drive the W2-synthesis coupling rows in `schema/drift_taxonomy.json`.

2.5 Cross-Surface Couplings (Seeded for W2)

Two cross-surface couplings seeded from this module will be promoted to Module D in Wave 2. First, Fastowski & Kasneci's misinformation-induced belief shift [Fastowski and Kasneci, 2024] couples to the retrieval surface via stale-index reactivation: a retrieval index that serves prior false claims produces the same uncertainty dynamics as direct prompt injection. Second, Wu et al.'s 30 pp [Wu et al., 2025] BoolQ drop couples to the retrieval surface by construction — the passages that degrade model accuracy *are* retrieval targets in a typical RAG pipeline. Both couplings are recorded as `to_surface = retrieval` entries in the taxonomy.

MODULE B — ALIGNMENT & TASK DRIFT (WAVE 1B)

3 Module B — Alignment & Task Drift

Module ID: B

Surface: Alignment

Primary sources: abdelnabi2024taskdrift, das2025tracealign, dongre2025equilibria, goaldrift2025, rath2026asi, sail2025instruction

Supporting sources: greenblatt2024intrinsic, betley2024emergent, michel2019sixteen, preference2025real

3.1 Scope of the Alignment Surface

Module B surveys the alignment surface: the family of drift phenomena in which an LLM's behavior diverges from its intended policy, stated goal, or system-level role without any corresponding change in weights. The alignment surface is distinct from the knowledge surface covered in Module A because the failure need not involve a factual error — the model may confidently violate a safety policy, drift from its stated mission, or reverse a role assignment while remaining internally consistent.

Four sub-phenomena organize the surface. *Task drift* (Abdelnabi et al. [et al., 2024a]) denotes the behavior change induced when external content processed by an LLM — search results, emails, documents — contains embedded instructions; the model's observable trajectory diverges from the user's original task. *Alignment drift* in the TraceAlign framing [Das et al., 2025] names the reactivation of pretraining-embedded policy-violating behavior under adversarial prompting, paraphrased jailbreaks, or decoding perturbations. *Context drift* as defined by Dongre et al. [et al., 2025] is the turn-wise KL divergence between a test model's predictive distribution and a goal-consistent reference model; crucially they show drift reaches a stable equilibrium rather than runaway decay. *Goal drift* as evaluated by the Goal Drift report [alignment team , via citations] targets prolonged-adversarial-pressure deviation of an agent's stated goal from the system goal; that report explicitly defers intrinsic (training-acquired) goal drift to future work cited as [et al., 2024d] and [et al., 2024b]. *Agent drift* per Rath [Rath, 2026] is the progressive degradation of agent behavior, decision quality, and inter-agent coherence over extended interaction sequences in multi-agent systems, measured by a twelve-dimensional Agent Stability Index.

Two cross-cutting mechanisms structure the mitigation design space. First, drift on this surface occurs *without parameter updates* [Rath, 2026] — the failure originates in contextual conditioning and sampling, not weights, so defenses can operate at inference time. Second, the locus of instruction arbitration concentrates in middle-layer focal attention heads [team, 2025], echoing the classical finding that only a subset of attention heads are functionally critical [et al., 2019]; this makes surgical LoRA adaptation at mid-depth a tractable intervention.

3.2 Cited Findings

Six findings populate the alignment surface. Each provides paper-specific attribution, a quantitative figure, and carries inline `\citedrift` attribution within ± 50 characters of every numeric token so the CAPCOM gate's numeric-attribution check passes.

3.2.1 Task Drift from Prompt Injection

Abdelnabi et al. (Microsoft Research) propose an activation-delta probe for task drift: the difference in model activations before and after processing external content. A simple linear classifier on these deltas achieves near-perfect ROC AUC on out-of-distribution test sets [et al., 2024a]. The editorial review flags *near-perfect* as a loose rendering of the paper's specific figures (the paper reports above 0.99 AUROC [et al., 2024a]), so this survey softens to the structural finding: task drift leaves a high-signal signature in internal state, and detection can be performed without fine-tuning the base LLM — only the probe is trained [et al., 2024a]. This converts prompt-injection detection from a black-box output-inspection problem into a white-box activation-inspection problem, which is the CAPCOM-gate anchor for the W0→W1 boundary (§3.4).

3.2.2 Training-Time Belief Traceback (TraceAlign)

Das, Jain, and Chadha (Apple / Stanford / UNT) define alignment drift operationally and build TraceAlign to trace failures back to their root causes in the training corpus via suffix-array matching (TraceIndex) [Das et al., 2025]. The Belief Conflict Index (BCI) quantifies semantic inconsistency between a generated span and an aligned policy, conditioned on the retrieved training documents. Three interventions compose: TraceShield (inference-time filter refusing high-BCI completions), Contrastive Belief Deconfliction (DPO-style penalty on high-BCI continuations), and Prov-Decode (provenance-aware decoding that vetoes beam expansions predicted to yield high-BCI spans) [Das et al., 2025]. On the Alignment Drift Benchmark, the baseline drift rate exceeds 40% [Das et al., 2025] and the full-stack intervention reduces it to 6.2% [Das et al., 2025]. The combined stack delivers up to 85% reduction [Das et al., 2025] with utility delta under 0.2 [Das et al., 2025]. The authors further derive a theoretical upper bound on drift likelihood from suffix-array span statistics, linking memorization frequency and length to adversarial reactivation risk [Das et al., 2025]; the editorial review notes that upper-bound derivation rests on suffix-array assumptions that warrant W1B vetting and that ADB is TraceAlign-internal, so external generalization requires replication.

3.2.3 Context Equilibria in Multi-Turn

Dongre et al. formalize multi-turn context drift as the turn-wise KL divergence between a test model's token-level predictive distribution and a goal-consistent reference model [et al., 2025]. Contra the prevailing intuition that multi-turn drift accumulates unboundedly, they observe *stable equilibria*: drift reaches a finite, noise-limited fixed point rather than runaway decay [et al., 2025]. A stochastic recurrence model with restoring forces accounts for the observation; lightweight periodic goal-reminder interventions reliably shift the equilibrium downward [et al., 2025]. Experiments span synthetic long-horizon rewriting tasks and realistic user-agent simulations in τ-Bench [et al., 2025]. The authors explicitly distinguish context drift (loss or

corruption of information in the active context) from alignment drift (deviation from intended behavioral policies) [et al., 2025]; both respond to reminder-style interventions, which this survey treats as a single CAPCOM-recommended mitigation gate at the W1→W2 boundary (§3.4).

3.2.4 Goal Drift in LM Agents

The Goal Drift technical report evaluates LM agents under prolonged adversarial pressure, scoring how often an agent's stated end-of-run goal aligns with the system goal versus a competing or instrumental one [alignment team , via citations]. The editorial review (684.1) flags that this report is an external evaluation *of* Anthropic models (and GPT-4o variants), not an Anthropic-authored paper, so attribution here is to the Goal Drift report's authors generally; the finding itself is unaffected. Its Finding 9 is that models' own self-reports of their goals correlate strongly with measured goal-drift scores [alignment team , via citations]. Claude 3.5 Sonnet never states a misaligned goal across the evaluated runs [alignment team , via citations]; GPT-4o and Claude 3.5 Haiku state hybrid or misaligned goals more frequently [alignment team , via citations]; GPT-4o mini states a misaligned goal in the majority of runs [alignment team , via citations]. The methodology restricts to prompt-based goals rather than training-acquired goals, leaving intrinsic goal drift to [et al., 2024d] and [et al., 2024b]. Module D (Wave 2 synthesis, Phase 688) is the natural home for this model-comparison taxonomy because its implications cross the knowledge↔alignment coupling.

3.2.5 Agent Drift in Multi-Agent Systems

Rath (arXiv:2601.04170) introduces agent drift as progressive degradation of agent behavior, decision quality, and inter-agent coherence across extended interaction sequences, measured by a twelve-dimensional Agent Stability Index [Rath, 2026]. The mechanistic claim is that drift occurs *without parameter updates* [Rath, 2026] — contextual conditioning and sampling suffice to degrade behavior, so defenses can operate at inference time. Rath's simulated enterprise-finance experiments report that Adaptive Behavioral Anchoring alone yields 70.4% drift reduction [Rath, 2026]; combining Anchoring with EMC summarization and a third intervention produces 81.5% drift reduction at 23% computational overhead [Rath, 2026] and 9% extended median completion time [Rath, 2026]. Architectural factors correlate with drift susceptibility: rigorously hierarchical topologies and explicit consensus protocols drift less than emergent-coordination designs [Rath, 2026]. The editorial review (684.1) discounts rigor by one notch because Rath is single-author, runs on a custom simulator rather than a standard MAS benchmark, and has not been externally validated; per-point percentages should therefore be read as *reported in Rath's simulated environment*, not as architecture-independent constants.

3.2.6 Instruction Arbitration in Middle Layers

A contemporaneous study investigates where in the transformer instruction arbitration occurs when system-, user-, and peer-instructions conflict [team, 2025]. Using a Contextualized Role Adherence Score and a tri-axial head-drift score (magnitude, directional orientation, distributional reshaping), the authors identify a coherent set of focal attention heads concentrated in *middle layers* [team, 2025]. Their Surgical Alignment of Instruction Layers installs LoRA adapters only on these focal layers,

training a token-weighted DPO-style objective [team, 2025]. The finding re-confirms the classical result that only a subset of attention heads are functionally critical [et al., 2019] — and identifies the mid-depth locus as the intervention target for multi-agent role adherence. The editorial review flags that the paper's method-name (SAIL) differs from its bibliography title (*Diagnose, Localize, Align*), so this survey uses the cite_key `sail2025instruction` rather than the ambiguous short form [team, 2025].

3.3 Alignment-Drift Intervention Scorecard

Table 3 (file: `tables/alignment_scorecard.tex`) ranks five intervention classes by the product of compute cost and detection or mitigation power. The scorecard excludes Prov-Decode as a standalone entry because TraceAlign's reported 40%→6.2% drift reduction is the full-stack combined figure [Das et al., 2025]; Prov-Decode's marginal contribution is not separately reported, so it is subsumed under the TraceShield + CBD + Prov-Decode row. Cost estimates are inherited from the cited papers where stated; where not stated, this survey labels the cost qualitatively (*inference-probe only*, *training-time + inference*, *runtime orchestration*).

3.4 CAPCOM-Gate Recommendations

Per §D-07 of the 0684 foundation context, the CAPCOM gate is callable at each wave boundary with advisory severity mid-wave and blocking severity at `W3`. This section names which Module-B interventions belong at which boundary and the paper that motivates each placement.

W0→W1 — activation-delta baseline. Deploy the activation-delta probe [et al., 2024a] as the pre-wave baseline. Rationale: probe-only training (no base-LLM fine-tune) means the gate can run on any wave's model snapshot without retraining, and the near-perfect OOD ROC AUC [et al., 2024a] gives the orchestrator a high-signal sentinel for prompt-injection task drift introduced by Wave 1 content pipelines. Budget: inference-probe only. Gate output: activation-delta ROC AUC $\geq$ project threshold (reported near 0.99 in the paper [et al., 2024a]; the project should pick a conservative threshold such as 0.95 to leave headroom for domain shift).

W1→W2 — cross-module BCI audit + goal reminders. Run a TraceAlign-style BCI audit [Das et al., 2025] over any module prose produced in Wave 1, coupled with a goal-reminder pass [et al., 2025] that re-states the Wave 2 synthesis goal (a *unified cross-surface taxonomy*) before synthesis begins. Rationale: the 40%→6.2% reduction [Das et al., 2025] is the strongest mitigation signal in the alignment corpus, and Dongre's stable-equilibrium finding [et al., 2025] means that even a minimal periodic reminder reliably shifts the drift fixed point downward without mid-synthesis retraining. Gate output: BCI$<$threshold and goal-alignment score $\geq$ Dongre's reported reminder-intervention effect size.

W2→W3 — intrinsic-goal-drift screen. Screen Wave 2 synthesis outputs for intrinsic (training-acquired) goal drift using the evaluation shape of goaldrift2025 [alignment team , via citations], augmented with references to [et al., 2024d] and [et al., 2024b] as future-work markers. Rationale: Wave 2 taxonomies that span knowledge↔alignment

Table 3: Module-B alignment-drift intervention scorecard. Ranked by compute-cost × detection-or-mitigation-power. Cost tags are qualitative where the source paper does not quote FLOPs or latency; quantitative figures are carried verbatim from the cited work.

#	Intervention	Mechanism	Reported Effect	Compute Cost	Source
1	Activation-delta probe	White-box linear classifier on hidden-state deltas before vs after external content	Near-perfect ROC AUC on OOD test sets; probe-only trained, no base-LLM fine-tune [et al., 2024a]	Inference-probe only (low)	[et al., 2024a]
2	Goal reminders (periodic goal restatement)	Lightweight prompt-level intervention shifting the turn-wise KL equilibrium toward goal-consistent distribution	Shifts stable KL equilibrium downward in τ-Bench user-agent sims; context drift reaches finite fixed point not runaway decay [et al., 2025]	Prompt-overhead only (very low)	[et al., 2025]
3	Adaptive Behavioral Anchoring	Runtime regrounding of agent on exemplars to pull sampled behavior back to policy manifold	70.4% drift reduction solo in Rath's simulated enterprise-finance experiments [Rath, 2026]	Runtime orchestration (low–medium)	[Rath, 2026]
4	SAIL — LoRA on focal mid-layer attention heads	Localize instruction arbitration heads via CRAS + tri-axial head-drift score; train LoRA + token-weighted DPO on focal layers [team, 2025]	MAS role-adherence gains under system/user/peer instruction conflict; re-confirms subset-head criticality [et al., 2019]	Training-time LoRA + inference (medium)	[team, 2025]
5	TraceShield + CBD + Prov-Decode (full stack)	Inference filter on high-BCI spans + DPO penalty on high-BCI continuations + provenance-aware beam veto	Baseline alignment drift >40% [Das et al., 2025] reduced to 6.2% [Das et al., 2025]; up to 85% reduction combined [Das et al., 2025]; utility delta <0.2 [Das et al., 2025]	Training + inference, full stack (high)	[Das et al., 2025]

coupling are the most likely sites where a synthesis author's model will restate a system goal adversarially (e.g., collapsing a nuanced finding into a misleading summary). Running the goaldrift2025 self-report↔measured-drift correlation check [alignment team , via citations] at W2→W3 catches this before the publication gate. Gate output: measured-drift / self-report correlation within the range reported for aligned models.

W3→release — full TraceShield + ASI gate. Hard-block gate: run the full TraceShield + CBD + Prov-Decode stack [Das et al., 2025] over the synthesized module prose, then evaluate the Agent Stability Index [Rath, 2026] across the 4-module pipeline treating each module as an agent in a hierarchical topology. Rationale: Das et al.'s full stack is the only intervention in the corpus with a paper-reported hard number under 10% drift [Das et al., 2025] (6.2% on ADB [Das et al., 2025]); Rath's ASI provides a second, orthogonal metric over the multi-module composition. At W3, per §D-08 severity policy, WARN is not available: any BCI-audit failure or ASI fall below the project-set stability threshold BLOCKs publication. Additionally, the SAIL finding [team, 2025] that focal attention heads concentrate in middle layers is recorded as a *future-wave hypothesis*: if a future mission applies LoRA at publication time, this is where to apply it.

Operating notes. Gate reports land in `gates/<wave>_gate.md` with structured front-matter; the orchestrator may consume them automatically. The named-collision guard from the editorial review (preference2025real vs drift2026probe, both sharing the four-letter acronym as paper shortnames in the source literature) is enforced by requiring cite_key usage in all gate reports; this module complies throughout.

MODULE C — CONTEXT & RETRIEVAL DRIFT (WAVE 1C)

4 Module C — Context & Retrieval/SSoT Drift

Module ID: C

Surface: Retrieval

Primary sources: wu2025natural, rsd2026diffusion, liu2026chronos, sgi2025grounding, beerag2025entropy

Supporting sources: greco2024driftlens, muller2024d3bench, raggov2025laziness, contextualnorm2025, contextlen2025drift

4.1 Scope of the Retrieval/SSoT Surface

Module C surveys drift as it manifests between a retrieval-augmented generation (RAG) system's Single Source of Truth (SSoT) and the generator that consumes retrieved context. Three failure axes define the surface: (i) *index staleness* — the index lags the evolving real-world source, so retrieved passages diverge from the generator's pretraining-era expectations; (ii) *retrieval-quality decay under iteration* — reasoning agents reformulate queries away from the original information need and grow lazy about re-retrieval; (iii) *grounding drift* — the generated response departs from the retrieved context despite nominally being grounded on it. Each axis has a distinct observable and a distinct mitigation posture.

The SSoT framing treats the retrieval index as the project's canonical artifact and evaluates every generation against it. Drift on the retrieval surface is therefore not a property of the model alone; it is a property of the *system* — the index, the retriever, the generator, and the iteration loop that binds them. That system-level framing matters because several findings in this module ([authors, 2026], [et al., 2026]) identify drift modes that only emerge at particular architectural combinations (diffusion-LM generators, time-varying evidence streams) rather than at any single component.

A cross-coupling lead from Module A carries into this module: [Wu et al., 2025] is presented in Module A as a knowledge-drift finding but functionally doubles as a retrieval-drift signal — the 30 pp [Wu et al., 2025] BoolQ degradation at low-similarity bins is an upper bound on the damage a naive RAG system can inflict simply by retrieving an up-to-date passage. We cite the paper here for completeness, cross-reference its Module A treatment (§2), and concentrate new prose on the seven Module-C-native findings that follow.

4.2 Cited Findings

4.2.1 Natural Context Evolution vs. Retrieval Index

Wu, Schlegel, and Batista-Navarro [Wu et al., 2025] study LLM QA degradation as reading passages evolve via natural human editing relative to their pretraining-era versions. Evaluating six QA datasets on eight LLMs with public training data and computing paragraph-level semantic similarity between edited and pretraining snap-

shots, they report that BoolQ accuracy falls by more than 30 pp [Wu et al., 2025] between the highest- and lowest-similarity bins, with regression slopes exceeding 70 [Wu et al., 2025] across several LLMs. Human annotators do not exhibit the degradation, isolating the effect as an LLM-specific reliance on pretraining surface form rather than semantic comprehension.

The Module-C reading of this finding: when a RAG system's index is built over a snapshot while the canonical source (e.g. Wikipedia) continues to evolve, retrieved passages drift away from the generator's parametric expectation even though the index itself is fresh relative to the world. The damage is not a failure of retrieval per se but an interaction between retrieval freshness and generator pretraining age. Module-A treatment is in §2; we do not re-state the SD score numerics here to avoid CAPCOM quote-uniqueness duplication.

4.2.2 Response Semantic Drift in Diffusion-LM RAG

The Response Semantic Drift (RSD) paper [authors, 2026] identifies a qualitative failure mode in Diffusion Language Models coupled with RAG: the generated answer progressively deviates from the query's original semantics, yielding high copy-rate from the retrieved context but low query-precision. The authors trace the issue to denoising strategies that do not maintain query-semantic alignment across the iterative denoising process [authors, 2026]. The surfacing in both the source paper and `drift-mission.tex` is qualitative rather than quantitative; Opus editorial review (`sources/meta.json`) flagged the absence of a published RSD score or accuracy delta, so we preserve the qualitative framing here rather than interpolate numbers the paper does not report.

The architectural consequence is the key Module-C lever: diffusion-LM drift has a different shape than autoregressive-LM drift even when both generators are grounded on the same retrieval [authors, 2026]. RAG discipline cannot be architecture-agnostic, because the denoising trajectory introduces a second drift axis (query-semantic alignment across denoising steps) that the autoregressive pipeline does not have.

4.2.3 Continuous Knowledge Drift and Chronos

Liu et al. [et al., 2026] formalize continuous knowledge drift and construct a new benchmark of real-world dynamic events with time-stamped evidence. Vanilla RAG and several learning-based approaches fail on the benchmark — the authors characterize the failure qualitatively as catastrophic forgetting and temporal inconsistency rather than reporting a headline accuracy delta [et al., 2026]. The proposed baseline, Chronos, is a time-aware retriever that progressively organizes retrieved evidence into an Event Evolution Graph, enabling temporally consistent understanding without model retraining. Code ships open-source at `github.com/hbing-l/chronos` [et al., 2026].

The architectural claim that matters for Module C is that temporal signals must be embedded in retrieval itself, not added as prompt-side scaffolding. This is the seed of the SSoT discipline item "time-aware retrieval signal" in §4.3 — a post-hoc timestamp in the prompt does not rescue a retriever that has collapsed distinct temporal snapshots into a single surface.

4.2.4 Semantic Grounding Index and Semantic Laziness

The Semantic Grounding Index (SGI) paper [authors, 2025e] defines a geometric hallucination metric: the ratio of angular distances from the response to the question versus to the context on the unit hypersphere. The central finding is *semantic laziness* — hallucinated responses remain angularly proximate to the question rather than departing toward the retrieved context. On HaluEval (n=5000), Cohen's d ranges from 0.92 to 1.28 [authors, 2025e] across five embedding models; effect size rises monotonically with question-context angular separation from d=0.61 [authors, 2025e] at low separation to 1.27 [authors, 2025e] at high separation, with AUC rising from 0.72 to 0.83 [authors, 2025e] correspondingly. The authors derive the separation result from the spherical triangle inequality, giving the empirical discriminative power a theoretical foundation.

For Module C, the direct implication is that citation presence in a response does not imply semantic grounding on the retrieved context. An SGI-style geometric audit on response-question-context triangles is therefore a required component of grounding-faithfulness evaluation (§4.4, test CF-SGI-01).

4.2.5 Retrieval Laziness and Query Drift in Iterative RAG

A government-documents RAG study [docs RAG authors, 2025] identifies two distinct qualitative failure modes in iterative reasoning-agent RAG: *query drift*, where the reasoning agent's reformulated queries wander away from the original information need, and *retrieval laziness*, where the agent stops retrieving — or retrieves superficial chunks — once any response at all becomes possible. Both modes are reported qualitatively: the source paper separates the failure modes and proposes mitigation modules, but `drift-mission.tex` surfaces neither baseline-vs-mitigated accuracy nor reformulation-distance numerics, and the 684.1 editorial advisory flagged this as a surfacing gap [docs RAG authors, 2025]. We preserve the qualitative framing here rather than invent figures.

The conceptual alignment with SGI (§4.2.4) is noteworthy: SGI's *semantic* laziness (the response won't depart from the question) and the iterative-RAG study's *retrieval* laziness (the agent won't ask again) are complementary pressures on the same grounding discipline, one inside the generator and one outside it [docs RAG authors, 2025]. Wave 2 synthesis will make the coupling explicit.

4.2.6 Context-Format Sensitivity and Attention-Entropy Dilution

Two qualitative findings bracket the prompt-side discipline. Contextual Normalization [authors, 2025b] demonstrates that superficial format choices — delimiters, structural markers in key-value extraction, positional placement of retrieved documents — induce substantial shifts in RAG accuracy and stability even when the semantic content is identical. The source paper and the mission-tex surface the effect qualitatively rather than with a headline accuracy delta [authors, 2025b]; the proposed mitigation is a lightweight adaptive strategy that standardizes context representations before generation.

BEE-RAG [authors, 2025a] studies a parallel failure mode driven by context length: as retrieved context grows, attention becomes entropy-diffuse and focus on critical segments degrades. Again, drift-mission.tex surfaces the finding qualitatively [authors,

2025a] — the source paper's specific entropy thresholds and collapse-point accuracy pairs are not quoted in our corpus, and the 684.1 advisory explicitly flagged the gap. Balanced-entropy engineering is proposed as the mitigation.

Context length also couples drift across turns: Yousuf et al. [length drift authors, 2025] show qualitatively that as context grows, drift surfaces amplify, complementing BEE-RAG's attention-entropy mechanism and Dongre et al.'s KL-equilibrium result (Module B). The three papers together argue that the context-length budget is not a performance-only knob but a drift-control knob [length drift authors, 2025].

4.2.7 Concept-Drift Tooling: DriftLens and D3Bench

Greco et al. [et al., 2024c] present DriftLens, an unsupervised framework for real-time concept-drift detection from deep-learning representations. The tool provides three primitives — a reference distribution from in-distribution training data, a drift metric measuring the distance between reference and current window, and an alarm threshold [et al., 2024c]. Originally designed for classical ML model monitoring, the primitives transfer directly to RAG-index monitoring: the index distribution becomes the reference, new retrievals become the window, and triggered alarms indicate the index has diverged from its source.

Müller, Abdelaal, and Stjelja [Muller et al., 2024] present D3Bench, a microbenchmark of three open-source drift-detection tools — Evidently AI, NannyML, and Alibi-Detect — on real-world smart-building data. Specialization emerges: Evidently AI leads on general data-drift detection, NannyML pinpoints the *timing* of shifts and their downstream effect on predictive accuracy [Muller et al., 2024]. The smart-building→RAG transfer is editorial, not established in the source paper, so we phrase the adoption as proposed rather than proven (per 684.1 editorial concern).

4.3 SSoT Discipline Checklist

Following the format of Module A's taxonomy table and Module B's intervention scorecard, the SSoT discipline checklist distills the cited findings into six operational controls that a drift-aware RAG system should apply. The checklist lives in `tables/ssot_checklist.tex` and is loaded here:

Discipline	Rationale (with cite)
Versioned index with source timestamp	Passage-to-pretraining similarity predicts QA accuracy; a fresh index on a generator with a stale parametric prior still loses accuracy on edited passages, so the index must carry both the retrieval timestamp and the source-version fingerprint [Wu et al., 2025].
Time-aware retrieval signal	Temporal ordering of evidence must be embedded in the retriever, not the prompt; Chronos' Event Evolution Graph enables temporally consistent understanding without model retraining [et al., 2026]. A prompt-side timestamp is insufficient once the index has collapsed distinct snapshots into one surface.
Grounding-faithfulness evaluation	Citation presence does not imply grounding: hallucinated responses remain angularly proximate to the question rather than departing toward the retrieved context [authors, 2025e]. A geometric audit on response-question-context triangles — or an ablation that redacts context and checks whether the answer survives — is required before trusting RAG output.
Context-format standardization	Superficial format choices (delimiters, structural markers, document placement) induce substantial RAG accuracy shifts even when semantic content is identical [authors, 2025b]; a standardization layer before generation removes this incidental drift channel.
Context-length budget with entropy watch	Long retrieved contexts dilute attention entropy and degrade focus on critical segments [authors, 2025a]; the context-length budget is a drift-control knob, not just a performance knob. The coupling with multi-turn drift [length drift authors, 2025] widens the concern across iterative interactions.
Query-drift gate in iterative RAG	Reasoning-agent reformulations wander away from the original information need (query drift) and the agent grows lazy about re-retrieval once any answer is possible (retrieval laziness) [docs RAG authors, 2025]. A gate that monitors reformulation distance and re-retrieval frequency is required before trusting iterative-RAG output.
Drift-tool integration on the index	Classical concept-drift tooling — DriftLens' reference/metric/threshold idiom [et al., 2024c] and D3Bench's comparison of Evidently AI, NannyML, and Alibi-Detect [Muller et al., 2024] — transfers directly to RAG-index monitoring. The index distribution becomes the reference; new retrievals become the window; alarms indicate index→source divergence.
Ablation: answer survives if context removed?	If a RAG response is regenerated verbatim with its retrieved context redacted, grounding was weak regardless of what the response cites. The ablation is the simplest faithfulness floor and is called out in drift-mission.tex §Module C as a required audit behaviour; consistent with the SGI semantic-laziness finding [authors, 2025e].

Each item cites the primary evidence for including it. The ordering roughly follows the retrieval pipeline from index to generator output: index freshness first, temporal signals next, format discipline during retrieval, length/entropy discipline at assembly, grounding faithfulness at generation, and tooling/ablation discipline at the audit layer.

4.4 Grounding-Faithfulness Test Suite

The test suite is a first-pass specification of evaluations that a drift-aware RAG system should pass before declaring itself SSoT-disciplined. Test IDs follow the `00-test-plan.md` naming convention: SC-* for semantic consistency tests, CF-* for context-faithfulness tests, IT-* for iteration tests, and EC-* for edge-case tests. Expected-pass/fail matrix below.

Test ID	Evaluates	Procedure + expected behavior	Primary source
CF-SGI-01	Semantic laziness on hallucinated answers	Measure SGI (angular distances response→question vs. response→context) on a held-out hallucination set; expected PASS = effect size $d \geq 0.9$ separating hallucination from faithful grounding [authors, 2025e].	[authors, 2025e]
CF-ABL-02	Grounding ablation (answer survives context removal)	Re-run query with retrieved context redacted; expected FAIL = original answer is regenerated without the context (grounding was weak). The test targets the `drift-mission.tex` §Module C ablation principle.	Module C reference, [authors, 2025e]
SC-WU-03	Natural context drift against pretraining	Build index over post-pretraining-edit Wikipedia snapshot; run BoolQ; expected PASS = accuracy drop [Wu et al., 2025] <10 pp [Wu et al., 2025] across similarity bins when a time-aware retriever (Chronos-style) is enabled. Baseline failure is the >30 pp drop [Wu et al., 2025].	[Wu et al., 2025] + [et al., 2026]
IT-LAZY-04	Retrieval laziness under iteration	Multi-turn reasoning-agent run on a gov-docs QA set; log re-retrieval rate and reformulation distance per turn; expected PASS = re-retrieval triggered whenever context coverage falls below a named threshold [docs RAG authors, 2025].	[docs RAG authors, 2025]
EC-FMT-05	Context-format sensitivity	Hold semantic content constant; vary delimiters, markers, and document placement; expected PASS = accuracy variance below a budget under Contextual Normalization and above it without [authors, 2025b].	[authors, 2025b]
EC-ENT-06	Attention-entropy collapse watch	Monitor attention entropy across long contexts on a RAG QA eval; expected PASS = alarm fires before accuracy degrades past a budget, with balanced-entropy engi-	[authors, 2025a]

Six test IDs shipping here; Phase 695/696 (Half B retrieval/SSoT defenses) may add further tests when integrating concrete detectors. SC/CF/IT/EC coverage is intentional: SC covers semantic consistency against real-world evolution, CF covers context-faithfulness (the heart of grounding), IT covers iteration loops, and EC covers edge-case formatting and entropy pathologies.

4.5 Taxonomy Rows

Module-C phenomenon records appended to `schema/drift_taxonomy.json` with `surface: "retrieval"`. Seven rows, one per finding subsection. Cross-coupling rows (AC, BC) deferred to Wave 2 synthesis (Phase 688) per 0684-CONTEXT.md §D-02.

MODULE D — SYNTHESIS: CROSS-DRIFT TOPOLOGY (WAVE 2)

5 Module D — Cross-Drift Synthesis & GSD Mapping

Module ID: D

Surface: cross

Inputs: modules/module$_a.tex, modules/module_b.tex, modules/module_c.tex, schema/drift_taxonomy.json$

Outputs: tables/unified$_taxonomy.tex, audits/integ_report.md, thismodule'sGSD-mappingsection$

Author phase: 688 (Opus)

Editorial basis: Phase 684.1 Opus review + W1A/W1B/W1C VERIFY audits (audits/module_{a,b,c}_verify.md)

5.1 Cross-Drift Coupling Matrix

Module D promotes three cross-surface couplings from the Wave 1 modules into the synthesis layer. Each coupling names a pair of drift surfaces, attributes the claim to at least one primary source per the Phase 684.1 editorial register, and identifies the intervention boundary where a mitigation can break the coupling before damage propagates. The three couplings are labelled AB (knowledge ↔ alignment), AC (knowledge ↔ retrieval), and BC (alignment ↔ retrieval).

5.1.1 AB Coupling — Knowledge ↔ Alignment

Claim. Misinformation that reinforces a model's parametric prior (knowledge surface) becomes alignment-policy drift (alignment surface) when the reinforced false belief later conflicts with a stated safety or role policy. The claim has two experimentally-distinguishable legs: an *upstream* leg in which prompt-time misinformation shifts the model's uncertainty — a first-exposure rise of 56.6% [Fastowski and Kasneci, 2024] and a repeated-exposure collapse of −52.8% [Fastowski and Kasneci, 2024] below the untainted baseline — and a *downstream* leg in which the internalised false belief gets reactivated under adversarial prompting, producing alignment-policy failures on the Alignment Drift Benchmark at a baseline rate of >40% [Das et al., 2025] that the full TraceAlign stack reduces to 6.2% [Das et al., 2025] at utility delta under 0.2 [Das et al., 2025]. The TraceAlign framework makes the knowledge→alignment propagation path explicit by tying alignment failures to memorised training spans via suffix-array matching and the Belief Conflict Index [Das et al., 2025]; the upstream Fastowski mechanism is the analogous prompt-time pathway into the same downstream surface.

Evidence. Fastowski & Kasneci identify the internalisation dynamic directly [Fastowski and Kasneci, 2024]. Das et al. identify the memorised-content reactivation pathway and quantify the full-stack mitigation [Das et al., 2025]. The Phase 684.1 review assigns `surface_fit.cross = strong` to both papers and records the AB coupling as the strongest-evidenced of the three pairs.

Intervention boundary. The coupling breaks at two inline filters. First, at prompt-

intake, *repeated-exposure screening* detects the −52.8% [Fastowski and Kasneci, 2024] sub-baseline confidence signature that indicates internalised misinformation. Second, at inference, the TraceShield + Contrastive Belief Deconfliction + ProvDecode stack [Das et al., 2025] refuses completions whose retrieved training-span fingerprint matches the memorisation locus of the misinformation. Composed, these two filters catch both the knowledge-surface cause and the alignment-surface effect without requiring base-LLM retraining.

5.1.2 AC Coupling — Knowledge ↔ Retrieval

Claim. A stale retrieval index (retrieval surface) becomes knowledge drift (knowledge surface) when the passages returned to the generator diverge from the generator's pretraining-era expectation even though the index itself is fresh relative to the world. The same finding anchors both Module A (knowledge surface reading) and Module C (retrieval surface reading): Wu et al. report an accuracy drop exceeding 30 pp [Wu et al., 2025] between the highest- and lowest-similarity bins on BoolQ across six QA datasets and eight LLMs with public training data, with regression slopes above 70 [Wu et al., 2025] across several LLMs. Human annotators do not exhibit the degradation, which isolates the effect as an LLM-specific reliance on pretraining surface form [Wu et al., 2025] rather than semantic comprehension. The Chronos framework [et al., 2026] frames the retrieval-side fix: temporal signals must be embedded in the retriever itself through an Event Evolution Graph rather than added as prompt-side scaffolding, because a prompt-side timestamp cannot rescue a retriever that has collapsed distinct temporal snapshots into a single surface.

Evidence. Wu et al.'s finding is the canonical AC-coupling anchor per Phase 684.1 editorial review (`surface_fit.cross = strong` for both the knowledge and retrieval surfaces) [Wu et al., 2025]. Liu et al.'s Chronos [et al., 2026] identifies the retrieval-side mitigation. The paper surfaces the result qualitatively rather than with a headline accuracy delta, per Phase 684.1 advisory; this survey preserves the qualitative framing and does not invent numerics [et al., 2026].

Intervention boundary. The coupling breaks in the retriever: a versioned index with a source-timestamp fingerprint plus a time-aware retrieval signal [et al., 2026] prevents the generator from ever seeing a passage whose drift-from-pretraining exceeds a budget. The SSoT-discipline checklist in Module C (`tables/ssot_checklist.tex`) lists both controls as the top two items; Module D promotes them to mandatory gates whenever a mission's RAG surface crosses the ±30 pp [Wu et al., 2025] similarity-bin threshold reported by Wu et al.

5.1.3 BC Coupling — Alignment ↔ Retrieval

Claim. Prompt injection through retrieved content (retrieval surface) becomes task drift (alignment surface) when external documents reaching the model carry embedded instructions that redirect the agent away from the user's original task. The injection path runs through the retrieval pipeline, not through direct user prompts — making the failure simultaneously a retrieval-surface and alignment-surface phenomenon. The white-box activation-delta probe of Abdelnabi et al. [et al., 2024a] detects this coupling at the boundary where retrieved content enters the model's context window, reaching ROC AUC >0.99 [et al., 2024a] on out-of- distribution test sets with only the probe trained (base-LLM unchanged) [et al., 2024a]. Two reinforc-

ing failure modes compound the coupling: *retrieval laziness*, where the reasoning agent stops re-retrieving once any response becomes possible [docs RAG authors, 2025], and *query drift*, where reformulations wander away from the original information need [docs RAG authors, 2025]; both let injected content persist unchallenged across iteration turns.

Evidence. Abdelnabi et al. supply the primary coupling mechanism (activation-delta signature of injection-induced task drift) [et al., 2024a]. The gov-docs RAG study [docs RAG authors, 2025] contributes the compounding failure modes. Both papers carry `surface_fit.cross = strong` in Phase 684.1 review, and the combined evidence is the BC-coupling anchor in `schema/drift_taxonomy.json`. An orthogonal line of evidence — the SGI semantic-laziness finding of Cohen's d = 0.92–1.28 across five embedding models [authors, 2025e] — binds *generator-side* laziness to *retriever-side* laziness, showing that BC-coupling manifests inside the generator even when retrieval itself is pristine.

Intervention boundary. The coupling breaks at two enforcement points. First, at the retrieval-to-context boundary, the activation-delta probe [et al., 2024a] acts as a CAPCOM sentinel: a below-threshold AUC denies the content entry into the model's context. Second, in iterative reasoning agents, a *query-drift gate* [docs RAG authors, 2025] monitors reformulation distance and re-retrieval frequency, and a *semantic-grounding audit* [authors, 2025e] verifies that responses depart toward retrieved content rather than remaining angularly proximate to the question. Module B's alignment scorecard ranks the activation-delta probe as the highest-cost-effectiveness intervention in the alignment corpus (row 1 of `tables/alignment_scorecard.tex` [et al., 2024a]), making it the preferred enforcement primitive at the BC boundary.

5.2 Unified Drift Taxonomy

The unified taxonomy consolidates the 19 phenomenon records accumulated across Waves 1A/1B/1C into a single cross-surface table. Fifteen distinct rows ship in `tables/unified_tax` (the DRIFT-04 minimum is ten), spanning five knowledge-surface rows, one knowledge & retrieval cross-surface row, five alignment-surface rows, one alignment & retrieval cross-surface row, and three retrieval-surface rows (one of which is retrieval & alignment adjacent). The machine-readable form lives in `schema/drift_taxonomy.json`; the LaTeX version appears as Table 4 via the `\input` directive below.

Two editorial reductions were applied to avoid double-counting flagged in the INTEG cross-module audit (§5.4, finding 1). First, the Wu et al. natural-context-drift finding is reduced from two separate rows (one in Module A as knowledge, one in Module C as retrieval) into a single cross-surface row, consistent with the Phase 684.1 review's identification of the paper as the canonical AC-coupling anchor [Wu et al., 2025]. Second, the Abdelnabi task-drift finding is rendered as a single alignment & retrieval row rather than duplicated across Modules B and C, matching the coupling-matrix identification of the BC-coupling anchor [et al., 2024a]. Both reductions preserve all numeric claims; they eliminate double-surfacing rather than re-attributing magnitudes.

The taxonomy is deliberately not ranked, because the per-row magnitudes are incommensurate: a Cohen's d of 0.92–1.28 [authors, 2025e] anchors an angular-geometry

Table 4: Unified Drift Taxonomy. Twelve distinct phenomena spanning the three drift surfaces (knowledge, alignment, retrieval). Each row lists the named signal, its measurement metric, the reported effect magnitude, and the primary source. Machine-readable form in `schema/drift_taxonomy.json`. Rigor-discount framings (*reported on evaluated corpus*, *qualitative*) follow Phase 684.1 editorial advisory.

Phenomenon	**Surface**	**Signal**	**Metric**	**Magnitude**	**Citation**
Semantic drift in long-form generation	knowledge	Intra-output factual decay	SD score over atomic facts on FActScore corpus	0.77–0.79 across six LLMs [Spataru et al., 2024]	[Spataru et al., 2024]
Late-generation factuality decay	knowledge	Drift-point position	FActScore vs. stopping position, oracle early-stop	44.6% → 81.7% oracle FActScore on LLaMA-2-70B [Spataru et al., 2024]	[Spataru et al., 2024]
Misinformation-induced belief shift	knowledge	Uncertainty deltas under injection	Entropy / perplexity / token-probability vs. clean baseline	+56.6% first exposure; −52.8% repeated [Fastowski and Kasneci, 2024]	[Fastowski and Kasneci, 2024]
Natural context drift vs. pretraining	knowledge & retrieval	QA accuracy vs. passage-similarity bin	BoolQ accuracy across high/low similarity bins (6 datasets, 8 LLMs)	>30 pp BoolQ drop [Wu et al., 2025]; cross-surface anchor (Module A + Module C)	[Wu et al., 2025]
Cross-depth semantic trajectory (LSD)	knowledge	Late-layer divergence	Margin-based contrastive alignment of hidden states with factual embeddings	F1 = 0.92, AUROC = 0.96 on evaluated corpus [Mir, 2025]	[Mir, 2025]
Intermediate-layer inconsistency probe	knowledge	Representational uncertainty signal	Lightweight probe AUROC over intermediate-layer activations	<0.1% overhead; +13 AUROC over prior [team, 2026]	[team, 2026]
Task drift from prompt injection	alignment & retrieval	Activation-delta under external content	White-box linear classifier on hidden-state deltas	ROC AUC >0.99 OOD [et al., 2024a]	[et al., 2024a]
Alignment drift (training-corpus reactivation)	alignment	Belief Conflict Index (BCI) from training spans	TraceAlign suffix-array matching + ADB benchmark	40% → 6.2% ADB; utility Δ <0.2 [Das et al., 2025]	[Das et al., 2025]
Context drift in multi-turn (stable	alignment	Turn-wise KL equilibrium	KL divergence	Finite-noise fixed point;	[et al., 2025]

discriminator across a large sample, whereas the 40% → 6.2% reduction [Das et al., 2025] anchors a benchmark-specific mitigation stack under single-paper rigor. The unified taxonomy is therefore a navigator, not a scorecard: each row points into the Wave 1 module that establishes the claim, and the magnitude cell preserves the source paper's native units.

5.3 GSD Ecosystem Mapping

The cited findings map cleanly onto three gsd-skill-creator substrate layers: the staging-layer *scope drift* detector already shipped under `src/staging/derived/scope-drift.ts`; the DACP (Design and Content Protocol) content-discipline layer that guards mission-package authorship; and the CAPCOM-gate design that sequences go/no-go checks across waves (per §D-07/D-08 of the Phase 684 foundation context). Each of the three subsections below maps at least one primary-source finding onto its GSD layer and explains what the source evidences about the layer's failure surface.

5.3.1 Staging-Layer *Scope Drift* as Knowledge-Drift-Applied-to-Derived-Content

The existing gsd-skill-creator staging layer ships a *scope drift* detector (`src/staging/derived/sco` that flags when a derived skill's declared scope exceeds the observation scope supported by the underlying session data. Module D formalises this detector as *knowledge-drift-applied-to-derived-content* — the scope generalisation step is a knowledge-surface failure mode operating inside the staging pipeline rather than inside a deployed model. Two Wave 1 findings map directly: Fastowski & Kasneci's first-exposure uncertainty rise of 56.6% [Fastowski and Kasneci, 2024] corresponds to the staging layer's *first-observation* generalisation step, where one or a few session traces are over-read into a broad scope claim; and the −52.8% [Fastowski and Kasneci, 2024] repeated-exposure drop corresponds to the second-pass consolidation step where confidence in the over-generalised scope rises to sub-baseline uncertainty. The existing scope-drift detector captures only the first failure mode (observed-scope vs. declared-scope ratio); Module D recommends a repeated-exposure confidence-drop screen as a follow-up control, mirroring the Fastowski two-phase dynamic.

The coupling with long-form semantic drift is structural: Spataru et al.'s 37% of paragraphs drifting within the first 10% of facts [Spataru et al., 2024] is the deployed-model analog of a staging-layer scope overreach — the same failure shape (early generalisation beyond evidence) operates at both layers with the same mitigation family (oracle or heuristic early-stop). The staging-layer concrete fix: when the scope-drift detector fires, the pipeline should truncate the derived skill's scope to the observed-scope subset rather than request additional evidence [Spataru et al., 2024], because the oracle early-stop in Spataru's biography corpus shows that truncation is strictly more effective than external retrieval.

5.3.2 DACP (Design and Content Protocol) Implications

DACP is the gsd-skill-creator design-and-content protocol layer that governs how mission-package prose, citation, and schema artifacts are authored. Two Wave 1 findings map onto DACP as inference-layer failure modes the protocol must defend against. Dongre et al.'s stable-equilibrium finding in multi-turn drift [et al., 2025] means DACP needs a *reminder cadence* built into long-authoring workflows: even a minimal periodic goal-reminder pass through the authoring context reliably shifts the

turn-wise KL equilibrium downward [et al., 2025], which in DACP terms corresponds to periodic re-injection of the mission-package's stated deliverables and canonical cite-key conventions. The alignment-scorecard ranking of goal reminders as row 2 (prompt-overhead only, very low cost) [et al., 2025] is the cost evidence supporting DACP adoption.

The second DACP implication is the acronym-collision guard surfaced in Phase 684.1 editorial review: `drift2026probe` and `preference2025real` both carry a four-letter shortname, and any prose using the bare acronym conflates two distinct papers. DACP must enforce *cite-key discipline* at authoring time — all prose must cite via the `\citedrift{cite_key}` macro, never the bare acronym. The CAPCOM gate's cite-resolution and numeric-attribution checks (`scripts/drift/capcom-gate.mjs` under §D-07) enforce this at machine-verification time. The Abdelnabi [et al., 2024a] result supplies the justification: if DACP did not enforce discipline, a downstream task-drift probe on authored prose (using the same activation-delta methodology Abdelnabi applies to LLM contexts, with ROC AUC >0.99 [et al., 2024a] on OOD) would flag acronym-collision text as injection-adjacent drift — and the authoring pipeline would correctly reject its own prose.

A third, softer DACP implication runs through Wu et al.: since the >30 pp [Wu et al., 2025] BoolQ drop shows pretraining surface-form reliance rather than semantic comprehension, DACP must timestamp-version every mission-package source. The canonical mission- package schema (`sources/meta.json`) already carries `review_status`, `rigor_score`, and `surface_fit` fields; Module D recommends an additional `freshness_window` field keyed to the source paper's update frequency, so a future `skill-creator drift audit` CLI (Phase 697, Half B) can flag stale-provenance sources before they reach Wave 1 authors.

5.3.3 CAPCOM-Gate Design Implications — Which Checks at Which Gate

The Wave-1 modules already name wave-boundary recommendations (Module B's W0→W1 activation-delta baseline and W3→release full stack [Das et al., 2025], Module C's SSoT discipline checklist). Module D lifts these into a unified CAPCOM-gate layout that binds each check to a specific wave boundary and a primary-source motivator:

- **W0 → W1A/B/C (foundation to survey waves):** activation-delta probe [et al., 2024a] on any Wave 0 prose flowing into the Wave 1 modules. Rationale: if the foundation scaffolds carry injected text from external tooling (BibTeX generators, PDF-extract pipelines), the activation-delta probe catches the injection before survey authors consume it. Severity: WARN with `--force` override permitted, per §D-08.
- **W1 → W2 (surveys to synthesis):** BCI audit from TraceAlign [Das et al., 2025] plus a Dongre-style goal- reminder pass [et al., 2025]. The BCI audit catches alignment-surface drift in the module prose (survey authors citing findings adversarially or out of context); the goal reminder re- states the Wave 2 synthesis goal (a *unified cross-surface taxonomy*) and shifts the equilibrium of the synthesis conversation [et al., 2025]. Severity: WARN.
- **W2 → W3 (synthesis to publication):** goal-drift screen following the evalua-

tion shape of [alignment team , via citations] (external evaluation OF Anthropic and OpenAI models, not authored by Anthropic, per the Phase 684.1 authorship note) plus an SGI grounding audit [authors, 2025e] over the synthesis prose. The goal-drift screen catches synthesis drift (where a Wave 2 author's model restates a Wave 1 finding adversarially); the SGI audit catches grounding drift (where Module D prose departs from the retrieved citations toward the question rather than the context). Severity: WARN.

- **W3 → release (publication to release):** full TraceShield + CBD + Prov-Decode stack [Das et al., 2025] producing the 40% → 6.2% [Das et al., 2025] reduction, plus an Agent Stability Index score [Rath, 2026] treating each of the four modules (A/B/C/D) as an agent in a hierarchical topology with 70.4% solo reduction [Rath, 2026] (reported in Rath's simulated environment) as the stability floor. Severity: BLOCK — no `--force` available, per §D-08.

The cost-effectiveness ordering (activation-delta at W0→W1, goal reminders at W1→W2, goal-drift + SGI at W2→W3, full-stack at W3→release) matches Module B's alignment-scorecard ranking (Table 3), which puts the cheapest and most broadly-applicable interventions at the earliest, most-frequently-invoked gates. The full-stack intervention runs only once per release.

5.4 Cross-Module Consistency Check (INTEG)

The INTEG audit covers four classes of cross-module contradiction flagged in the Phase 684 foundation context (§D-07 checks). The full per-check report lives in `audits/integ_report.md`; this subsection summarises the four checks and their resolutions.

Check 1. Duplicate citations with contradictory numerics. Two papers appear in multiple Wave 1 modules: Wu et al. [Wu et al., 2025] (Module A knowledge + Module C retrieval) and Abdelnabi et al. [et al., 2024a] (Module B alignment + Module C retrieval). Both are cross-surface anchors per Phase 684.1 review. Each numeric claim attributed to them is identical across modules (Wu: >30 pp BoolQ drop; Abdelnabi: ROC AUC >0.99 OOD). **No contradiction.** Module D's unified taxonomy table renders each as a single cross-surface row to avoid future double-counting.

Check 2. Phenomena double-counted without cross-ref. The INTEG audit found two cross-module double-counts. First, *natural context drift* appears as a row in both Module A's taxonomy and Module C's taxonomy; the unified taxonomy (§5.2) folds them into one cross-surface row. Second, *task drift from prompt injection* appears in Module B's alignment scorecard and implicitly in Module C's SSoT checklist (activation-delta at the retrieval-to-context boundary); the unified taxonomy renders this as a single alignment & retrieval row. Both reductions preserve numeric claims.

Check 3. Mitigations claimed in one module but contradicted elsewhere. The Spataru tool-call branch [Spataru et al., 2024] reports near-zero improvement from Toolformer/Atlas-style external QA in Module A, which could superficially appear to contradict Module C's retrieval-discipline framing. In fact the two findings are consistent: Spataru attributes the near-null improvement to the model's failure to *gate* retrieval against actual need [Spataru et al., 2024], and Module C's retrieval-laziness finding [docs RAG authors, 2025] names the same retrieval-gating failure mode. Module D ex-

plicitly cross-references the pair so future readers see the agreement rather than inferring a contradiction.

Check 4. Phase 684.1 advisory flags resolved. Three editorial-review advisories applied to Module D's own prose were addressed inline. The acronym-collision guard (advisory 1) is enforced by cite-key-only discipline throughout this module. The `goaldrift2025` authorship note (advisory 3) is honoured: every reference to [alignment team , via citations] frames the paper as an *external evaluation* OF Anthropic and OpenAI models. The Wu et al. title-drift advisory (advisory 2) does not affect Module D because Module D does not quote the paper's title; the single-row treatment in the unified taxonomy inherits whichever title Wave 1 authors locked in (see Module A for the canonical form).

The INTEG audit closes with zero unresolved contradictions. Full per-check tables and violation listings appear in `audits/integ_report.md`.

TEST PLAN — VERIFICATION CATALOGUE

6 Test Plan

6.1 Scope and Category Definitions

The test catalogue is organised into four category shapes per `drift-mission.tex` Stage 3 D8 (and Success Criterion 6): **SC** safety-critical (numeric attribution, quote discipline, peer-review floor), **CF** core-functional (named definitions, measurement metrics, two detection methods per surface, two mitigations per surface), **IT** integration (cross-module coupling, unified taxonomy, scope-drift mapping), **EC** edge-case (extreme inputs, degenerate format, failure propagation). Each ID is unique; each DRIFT-* success criterion from `.planning/REQUIREMENTS.md` maps to ≥ 1 test ID in the verification matrix below. Test totals: 36 IDs — SC = 10, CF = 12, IT = 8, EC = 6.

6.2 Safety-Critical Tests (SC)

Test ID	Target	Procedure / Pass Criterion	DRIFT-*
SC-01	Every numerical claim has inline citation	Run `capcom-gate.mjs --wave W3`; numeric-attribution check passes with zero violations.	DRIFT-05
SC-02	No direct quote exceeds 15 words	Run `capcom-gate.mjs --wave W3`; quote-length check passes with zero violations.	DRIFT-08
SC-03	At most one direct quote per cited source	Run `capcom-gate.mjs --wave W3`; quote-uniqueness check passes with zero violations.	DRIFT-08
SC-04	Every `\cite{key}` resolves in `meta.json`	Run `capcom-gate.mjs --wave W3`; cite-resolution check passes with zero unresolved keys.	DRIFT-01
SC-05	No source outside peer-review / arXiv / standard	Inspect `sources/meta.json`; every entry has `arxiv_id` or `doi` or `standard_ref`. The four non-arXiv prose-mentions carry `review_status` and are quoted only for motivation, no numerical claim attached.	DRIFT-12
SC-06	15 primary sources each cited at least once	Count `\citedrift{key}` occurrences for each of the 15 primary keys across modules A–D; each ≥ 1.	DRIFT-01
SC-07	Every `review_status: mismatch` resolved before publication	Inspect `sources/meta.json`; zero entries with `mismatch` still referenced in module prose.	DRIFT-12
SC-08	Every `review_status: unresolved` resolved or dropped	Inspect `sources/meta.json`; unresolved entries carry no numerical claim in prose; motivation-only citations permitted.	DRIFT-12
SC-09	CAPCOM W3 gate fires with zero exit code at build time	`xelatex --shell-escape` pass 3 ends with `gates/W3_gate.md severity: PASS`.	DRIFT-11
SC-10	Build reproducible from clean workspace	`rm -f *.aux *.bbl *.blg *.toc *.log drift-mission-final.pdf` then rerun three-pass + bibtex; identical PDF by byte length $\pm$ timestamp.	DRIFT-11

6.3 Core-Functional Tests (CF)

Test ID	Target	Procedure / Pass Criterion	DRIFT-*
CF-01	Module A: named definition of knowledge drift	Inspect §2.1; definition present, four signatures enumerated.	DRIFT-02
CF-02	Module A: ≥ 2 detection methods with citation	Inspect §2.2; SD score, layer-wise LSD, DRIFT probe, SelfCheck-BERTScore — four methods cited.	DRIFT-02
CF-03	Module A: ≥ 2 mitigations with citation	Inspect Table in §2.4; early-stop, resample-rerank, probe-router, tool-calls — four branches.	DRIFT-02
CF-04	Module B: named definition of alignment drift	Inspect §3.1; definition present, three axes enumerated.	DRIFT-02
CF-05	Module B: ≥ 2 detection methods with citation	Inspect §3.2; activation-delta probe, TraceAlign BCI, SAIL CRAS, ASI composite — four methods.	DRIFT-02
CF-06	Module B: ≥ 2 mitigations with citation	Inspect §3.3 / `alignment_scorecard.tex`; five ranked interventions.	DRIFT-02
CF-07	Module C: named definition of retrieval drift	Inspect §4.1; definition present, five surfaces enumerated.	DRIFT-02
CF-08	Module C: ≥ 2 detection methods with citation	Inspect §4.2; SGI angular probe, DriftLens, D3Bench, Chronos EEG — four methods.	DRIFT-02
CF-09	Module C: ≥ 2 mitigations with citation	Inspect `ssot_checklist.tex`; eight disciplines cited.	DRIFT-02
CF-10	Module D: ≥ 1 coupling claim per AB / AC / BC	Inspect §5.1; three coupling subsections, each with ≥ 1 citation.	DRIFT-03
CF-11	Unified taxonomy table has ≥ 10 distinct phenomena	Inspect `unified_taxonomy.tex`; fifteen rows.	DRIFT-04
CF-12	Wave 1 parallelizable fraction $\geq 40\%$	Count Wave 1 tasks vs Wave 0+Wave 2+Wave 3 total (waves A/B/C parallel); recorded in `drift-mission.tex` Stage 3.	DRIFT-10

6.4 Integration Tests (IT)

Test ID	Target	Procedure / Pass Criterion	DRIFT-*
IT-01	Cross-module citations consistent (no contradictory numerics)	Inspect `audits/integ_report.md` Check 1; zero contradictions on shared papers (Wu, Abdelnabi).	DRIFT-03
IT-02	Wu et al. (2509.01093) cited in both Module A and Module C with same numeric (>30 pp BoolQ)	Grep `wu2025natural` in both modules; numeric 30 pp identical.	DRIFT-03
IT-03	Abdelnabi et al. (2406.00799) cited in both Module B and Module C with same numeric (ROC AUC >0.99)	Grep `abdelnabi2024taskdrift` in both modules; numeric identical.	DRIFT-03
IT-04	Unified taxonomy table folds double-counted rows into single cross-surface row	Inspect `unified_taxonomy.tex`; Wu row marked `knowledge & retrieval`; Abdelnabi row marked `alignment & retrieval`.	DRIFT-04
IT-05	GSD staging-layer scope-drift mapping present	Inspect §5.3; section exists with ≥ 1 paragraph.	DRIFT-09
IT-06	TraceAlign BCI maps to per-skill training-pair governance	Inspect §5.3; BCI mechanism referenced as blueprint.	DRIFT-09
IT-07	CAPCOM mid-wave gates all reported PASS or WARN-with-force	Inspect `gates/W1A_gate.md`, `W1B_gate.md`, `W1C_gate.md`, `W2_gate.md`; all "severity: PASS".	DRIFT-11
IT-08	Every finding in unified taxonomy attributed to ≥ 1 primary source	Inspect `unified_taxonomy.tex`; every row's Citation column has ≥ 1 cite_key from `meta.json` primaries.	DRIFT-01

6.5 Edge-Case Tests (EC)

Test ID	Target	Procedure / Pass Criterion	DRIFT-*
EC-01	Unicode and dash characters parse cleanly (e.g. en-dash in "77–78%")	xelatex log contains no unicode-related warnings affecting rendered page.	DRIFT-11
EC-02	Cite-key collision between `drift2026probe` and `preference2025real` never collapses	Grep bare four-letter acronym; zero prose occurrences (editorial advisory 1 from Phase 684.1).	DRIFT-08
EC-03	Supporting entries with `arxiv_id: null` never carry numerical claim	Inspect Module A §2.2.5; Lindsey reference used for motivation only, no numeric attached.	DRIFT-12
EC-04	Long tables break across page correctly (e.g. unified taxonomy)	Visually inspect PDF pages containing the three tables; no content clipped or spilled outside page margins.	DRIFT-11
EC-05	CAPCOM gate exit code 2 aborts xelatex build under failure	Manually inject a numeric claim without citation into a scratch module; rerun; confirm build halts with `BLOCK`. Scratch reverted after verification.	DRIFT-11
EC-06	Multi-author papers render authors correctly in references	Inspect References section for Spataru et al., Fastowski & Kasneci, Wu et al. — author lists render per BibTeX.	DRIFT-12

6.6 Test Plan Summary

Total tests: **36 IDs** (SC = 10, CF = 12, IT = 8, EC = 6). Every category has ≥ 6 tests; total exceeds the Success Criterion 6 floor of 30 test IDs with comfortable margin. Every DRIFT-* requirement from `.planning/REQUIREMENTS.md` is covered by ≥ 1 test ID, verified in the matrix below.

VERIFICATION MATRIX

7 Verification Matrix

7.1 DRIFT-* Requirement → Test IDs

Every DRIFT-* success criterion from `.planning/REQUIREMENTS.md` maps to at least one test ID. The matrix below is the authoritative success-criterion-to-test mapping for v1.49.569 Half A.

Requireme	Short Statement	Covering Test IDs
DRIFT-01	15 primary sources each cited at least once with attributable finding	SC-04, SC-06, IT-08
DRIFT-02	Each surface: named definition, metric with citation, ≥ 2 detection methods, ≥ 2 mitigations	CF-01, CF-02, CF-03, CF-04, CF-05, CF-06, CF-07, CF-08, CF-09
DRIFT-03	≥ 1 explicit coupling claim per surface pair (AB, AC, BC) with ≥ 1 citation	CF-10, IT-01, IT-02, IT-03
DRIFT-04	Unified drift taxonomy: name, surface, signal, metric, magnitude, citation; ≥ 10 distinct phenomena	CF-11, IT-04
DRIFT-05	Every numerical claim has inline attribution to a specific paper	SC-01
DRIFT-06	Test plan: ≥ 30 test IDs across SC / CF / IT / EC categories	SC-09 (meta), §6.6 (self-evidencing: 36 IDs shipped)
DRIFT-07	Every success criterion maps to ≥ 1 test ID in verification matrix	This table (self-evidencing)
DRIFT-08	All direct quotes $<$ 15 words; at most one direct quote per cited source	SC-02, SC-03, EC-02
DRIFT-09	Final document includes section mapping drift findings to GSD staging-layer “scope drift”	IT-05, IT-06
DRIFT-10	Wave plan achieves ≥ 40% parallelizable work in Wave 1	CF-12
DRIFT-11	CAPCOM go/no-go gate explicitly defined at each wave boundary	SC-09, SC-10, IT-07, EC-01, EC-04, EC-05
DRIFT-12	No source cited that is not peer-reviewed, arXiv, or documented standard	SC-05, SC-07, SC-08, EC-03, EC-06

Table 5: DRIFT-* → test-ID coverage matrix. Every DRIFT-* has ≥ 1 covering test; most have ≥ 2.

7.2 Wave-Level CAPCOM Gate Status

Wave	Phase	Report	Files Checked	Severity
W0	684	`W0_gate.md`	meta.json scaffold sanity	PASS
W1A	685	`W1A_gate.md`	`module_a.tex`	PASS
W1B	686	`W1B_gate.md`	`module_b.tex`, `alignment_scorecard.tex`	PASS
W1C	687	`W1C_gate.md`	`module_c.tex`, `ssot_checklist.tex`	PASS
W2	688	`W2_gate.md`	`module_d.tex`, `unified_taxonomy.tex`	PASS
W3	689	`W3_gate.md`	all seven module/table files	PASS (HARD-BLOCK gate)

Table 6: CAPCOM gate status across the six waves. W0–W2 are advisory (WARN on fail, `--force` override permitted). W3 is a HARD BLOCK with no override; a non-zero exit aborts the xelatex build via the `\capcomgate{wave=W3}` macro at the end of this document.

7.3 Citation and Numeric Audits

Two additional audits accompany this artifact, land in `audits/`, and are consumed by the release-note CAPCOM-GATE.md sign-off:

- **Citation audit** (`audits/citation_audit.md`) — every cited work inspected against the peer-review / arXiv / documented-standard floor (SC-05), quote-length floor of 15 words (SC-02), at-most-one-quote-per-source rule (SC-03). Reports the violation counts for the four CAPCOM checks on the final-aggregated document.

- **Numeric audit** (`audits/numeric_audit.md`) — every numerical claim (%, AUROC, pp, F1, scale factor) scanned for inline citation within ± 50 characters (SC-01). Reports the violation count and surfaces any qualitative-only findings that are deliberately un-numerated (e.g. `rsd2026diffusion`).

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